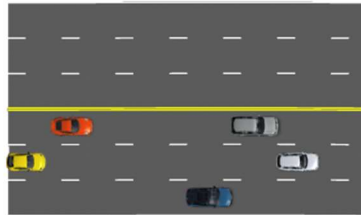


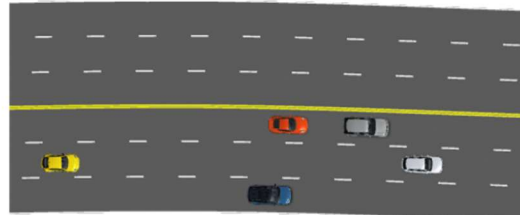
Scenario Definition – Sc-01

Sc-01: Lane Change to the Right - Infront of the Ego car		
Category	US101 Highway	
Type	Lane Change	
Manoeuvre	Lane change to the Right	
Scene	Dynamic Elements	Ego vehicle (Yellow), Target car: T00 (red Porsche 911), Lead car: T01 (white Audi A4), Surround vehicles: T02 (grey Range Rover), T03 (blue Nio ES8 SUV)
	Scenario	Dual carriageway with three lanes on each driving direction. Predominantly straight road segment with dashed line lane marking, Normal daylight conditions, Flat road conditions
	Self Representation	Ego vehicle - self driven with the IPG driver model, set to consider traffic, equipped with a radar and camera sensors, Target car - Longitudinal and lateral motions controlled manually, Surrounding vehicles - Lane keeping with longitudinal motion controlled manually.
Actions and Events	Initial scene at 0 sec: Ego car at the start position (0m) on R1 , Target car T00 at +5m from the start position on R0, Lead car T01 at +30m from start position on R1, T02 at +25m from start position on R0, T03 at +20m from start position on R2 Action 1- Target car accelerates Action 2 - Target car T00 initiates lane change --> Indicate Action 3 - Lane change Action 4 - Target car T00 completes the lane change from R0 to R1 Action 5 - Target car passes T02 to its left lane	
Goals & Values	Ego car to correctly identify and respond to the lane change initiated by the target vehicle and continue driving on R1 without any collisions.	
Description	The scenes in the scenario are illustrated in bird's-eye view. Forward direction is from the left to right on the illustration. The scenario consists of five vehicles driving along a highway of three lanes on each side. The self-driving (L4 HWP) ego vehicle is on lane R1 and on the lane to its left, the target vehicle (T00) and a surrounding car (T02) is moving forward. At the final scene, following the lane change by the target vehicle T00 (scene 5) , two leading cars (T00 & T01) are driving on the same lane (R1) as the ego vehicle. Scene 1: Initial scene illustrating the setting of the vehicles, their starting positions and the road layout for the scenario. Scene 1 --> Action 1: Target car accelerates --> Scene 2 Scene 2: The gap between the target car and its lead car (T02) narrows. Scene 2 --> Action 2: Indicate --> Scene 3 Scene 3: Target car T00 has started indicating its intention to change lane to R1. Scene 3 --> Action 3: Lane change --> Scene 4 Scene 4: Target car T00 is crossing the lane markings between R0 and R1, making a lane change manoeuver in front of the ego car. Scene 4 --> Action 4: Lane change completion--> Scene 5 Scene 5: Target car has completed the lane change manoeuver. Scene 5 --> Action 5 --> Scene 6 Scene 6: Final scene of all vehicles following their lanes with two cars leading the ego vehicle.	

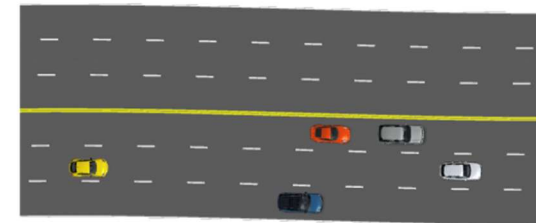
Sc-01 Representation



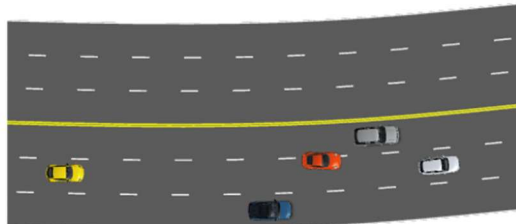
Scene 1



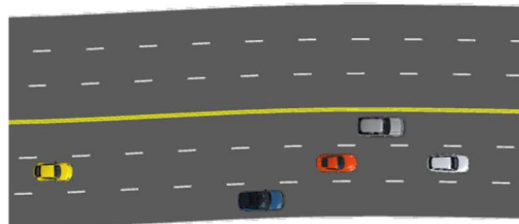
Scene 2



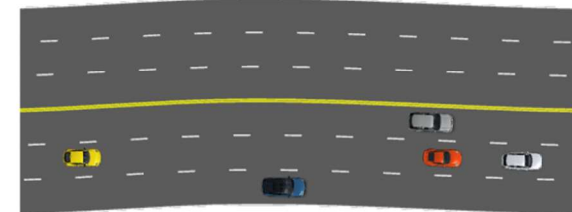
Scene 3



Scene 4

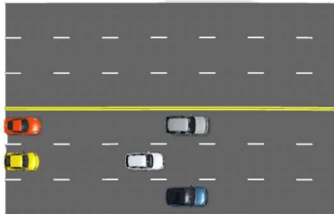
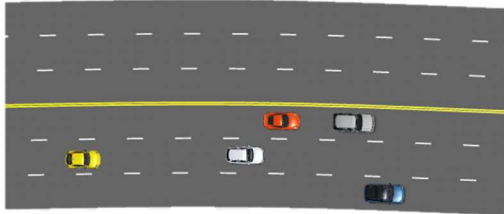
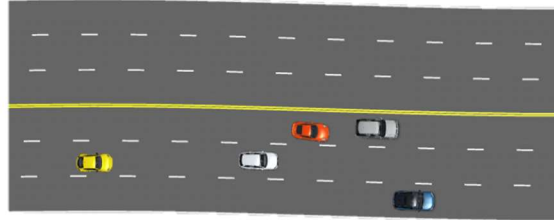
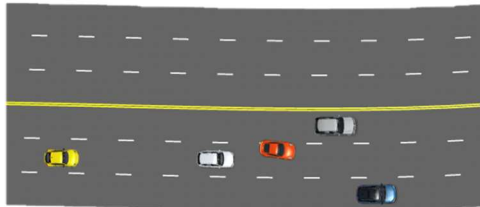
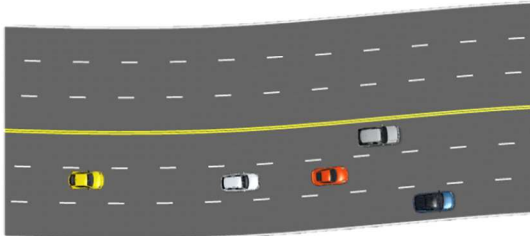
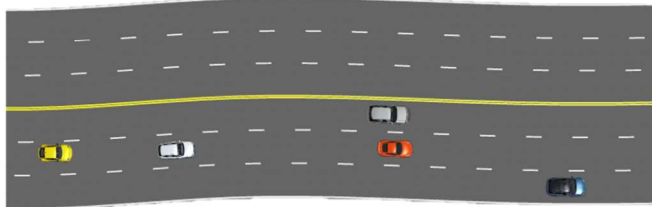


Scene 5



Scene 6

Scenario Definition – Sc-03

Sc-03: Lane Change to the Right - Infront of the Lead car		
Category		US101 Highway
Type		Lane Change
Manoeuvre		Lane change to the Right
Scene	Dynamic Elements	Ego vehicle (Yellow), Target car: T00 (red Porsche_911), Lead car: T01 (white Audi A4), Surround vehicles: T02 (grey Range Rover), T03 (blue Nio ES8 SUV)
	Scenariy	Dual carriageway with three lanes on each driving direction. Predominantly straight road segment with dashed line lane marking, Normal daylight conditions, Flat road conditions
	Self Representation	Ego vehicle - self driven with the IPG driver model, set to consider traffic, equipped with a radar and camera sensors, Target car - Longitudinal and lateral motions controlled manually, Surrounding vehicles - Lane keeping with longitudinal motion controlled manually.
Actions and Events		Initial scene at 0 sec: Ego car at the start position (0m) on R1 , Target car: T00 at the start position on R0, Lead car T01 at +15m from start position on R1, T02 at +20m from start position on R0, T03 at +20m from start position on R2 Action 1- Target car accelerates, passes the lead car Action 2 - Target car T00 initiates lane change --> Indicate Action 3 - Lane change Action 4 - Target car T00 completes the lane change from R0 to R1 Action 5 - Target car passes T02 to its left lane
Goals & Values		Ego car to correctly identify and respond to the lane change initiated by the target vehicle and continue driving on R1 without any collisions.
Description		The scenes in the scenario are illustrated in bird's-eye view. Forward direction is from the left to right on the illustration. The scenario consists of five vehicles driving along a highway of three lanes on each side. The self-driving (L4 HWP) ego vehicle is on lane R1 and on the lane to its left, the target vehicle T00 and a surrounding car (T02) is moving forward. At the final scene, following the lane change by the target vehicle T00 (scene 4) , two leading cars (T00 & T01) are driving on the same lane (R1) as the ego vehicle. Scene 1: Initial scene illustrating the setting of the vehicles, their starting positions and the road layout for the scenario. Scene 1 --> Action 1: Passes the lead car --> Scene 2 Scene 2: The gap between the target car and its lead car (T02) narrows. Scene 2 --> Action 2: Indicate --> Scene 3 Scene 3: Target car T00 has started indicating its intention to change lane to R1. Scene 3 --> Action 3: Lane change --> Scene 4 Scene 4: Target car T00 is crossing the lane markings between R0 and R1, making an unsafe cut in (lane change) manoeuver in front of the lead car. Scene 4 --> Action 4: Lane change completion--> Scene 5 Scene 5: Target car has completed the Cut-in manoeuver. Scene 5 --> Action 5 --> Scene 6 Scene 6: Final scene of all vehicles following their lanes with two cars leading the ego vehicle.
Sc-03 Representation		
 Scene 1  Scene 2  Scene 3  Scene 4  Scene 5  Scene 6		